

❖ HANDS-ON 9

❖ State Management — NgRx Store, Actions, Reducers, Effects & Selectors

The image displays a web application interface on the left and its Redux state on the right.

Web Application Interface (Left):

- Student Course Portal** (Navigation: Home, Courses, Profile, Enroll, Reactive Enroll)
- Courses**
- Angular**
- Course ID:** 2
- Course Code:** CS201
- Credits:** 3
- Status:** failed
- Enrolled Students**
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- [Back to Courses](#)

Redux State (Right):

- @ngrx/store/init** (1:43:19.84)
- @ngrx/effects/init** (+30:00.00)
- [Course] Load Courses** (+30:00.01)
- [Course] Load Courses** (+30:00.24) Success
- State:**
 - courses** (array):
 - 0 (object): { id: "1", name: "Java Progr...", code: "CS1..." }
 - 1 (object): { id: "2", name: "Angular", code: "CS201" }
 - 2 (object): { id: "4", name: "Database S...", code: "CS4..." }
 - 3 (object): { id: "5", name: "Operating ...", code: "CS5..." }
 - loading** (boolean): false
 - error** (string): null
 - enrollment** (object):
 - enrolledCourseIds** (array): []

Code Editor (Bottom):

The code editor shows the `course.reducer.ts` file. The code defines the initial state and the reducer logic for the course state.

```
1 import { createReducer, on } from '@ngrx/store';
2 import { Course } from '../models/course.model';
3 import {
4   loadCourses,
5   loadCoursesFailure,
6   loadCoursesSuccess,
7 } from './course.actions';
8
9 export interface CourseState {
10   courses: Course[];
11   loading: boolean;
12   error: string | null;
13 }
14
15 export const initialCourseState: CourseState = {
16   courses: [],
17   loading: false,
18   error: null,
19 };
20
21 export const courseReducer = createReducer(
22   initialCourseState,
23   on(loadCourses, (state) => ({
24     ...state,
25     loading: true,
26     error: null,
27   })),
28   on(loadCoursesSuccess, (state, { courses }) => ({
29     ...state,
30     courses,
31     loading: false,
32     error: null,
33   })),
34   on(loadCoursesFailure, (state, { error }) => ({
35     ...state,
36     loading: false,
37     error,
38   })),
39 );
```

TS course.actions.ts U X TS course.reducer.ts U TS course.effects.ts U TS enroll

```
student-course-portal > src > app > store > course > TS course.actions.ts > ...
1 import { createAction, props } from '@ngrx/store';
2 import { Course } from '../../models/course.model';
3
4 export const loadCourses = createAction('[Course] Load Courses');
5
6 export const loadCoursesSuccess = createAction(
7   '[Course] Load Courses Success',
8   props<{ courses: Course[] }>()
9 );
10
11 export const loadCoursesFailure = createAction(
12   '[Course] Load Courses Failure',
13   props<{ error: string }>()
14 );
```

TS app.config.ts M X TS app.ts M <> app.html M # app.css M TS course.reducer.ts U TS course.effects.ts U

```
student-course-portal > src > app > TS app.config.ts > ...
1 import { ApplicationConfig, provideBrowserGlobalErrorListeners } from '@angular/core';
2 import { provideRouter } from '@angular/router';
3
4 import { routes } from './app.routes';
5 import {
6   provideHttpClient,
7   withInterceptors
8 } from '@angular/common/http';
9
10 import { authInterceptor } from './interceptors/auth-interceptor';
11 import { errorHandlerInterceptor } from './interceptors/error-handler-interceptor';
12 import { loadingInterceptor } from './interceptors/loading-interceptor';
13 import { provideStore } from '@ngrx/store';
14 import { provideEffects } from '@ngrx/effects';
15 import { provideStoreDevtools } from '@ngrx/store-devtools';
16 import { courseReducer } from './store/course/course.reducer';
17 import { CourseEffects } from './store/course/course.effects';
18 import { enrollmentReducer } from './store/enrollment/enrollment.reducer';
19
20 export const appConfig: ApplicationConfig = {
21   providers: [
22     provideBrowserGlobalErrorListeners(),
23     provideRouter(routes),
24     provideHttpClient(
25       withInterceptors([
26         authInterceptor,
27         errorHandlerInterceptor,
28         loadingInterceptor
29       ])
30     ),
31     provideStore({
32       course: courseReducer,
33       enrollment: enrollmentReducer
34     }),
35     provideEffects(CourseEffects),
36     provideStoreDevtools({
37       maxAge: 25
38     })
39   ]
40 };
41
```